

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 12: Inertial Reference Systems (IRS/INS) & Schuler Tuning

Syllabus Source: CAE Oxford (pp. 453-498) | Pass Target: $\geq 90\%$ AviationExam

1. STRAPDOWN INERTIAL REFERENCE SYSTEMS (IRS)

Modern airliners use 'strapdown' IRS units with 3 orthogonal Ring Laser Gyroscopes (RLGs) and 3 quartz accelerometers fixed directly to the aircraft airframe:

- Ring Laser Gyro (RLG): Measures angular rate of rotation via the Sagnac Effect (interference between two counter-rotating laser beams). No moving mechanical parts, highly reliable.
- Accelerometer Integration: Accelerometers measure raw linear acceleration (a). 1st integration yields VELOCITY ($v = \int a \, dt$); 2nd integration yields POSITION ($s = \int v \, dt$).
- Gyrocompassing Alignment: On the ground, while stationary for 5 to 10 minutes, the IRS senses the Earth's rotation ($15^\circ/\text{hour}$) to automatically find TRUE NORTH and calculate local LATITUDE without external aid! Longitude MUST be manually entered by the pilot.

2. THE SCHULER TUNING PRINCIPLE (84.4 MINUTE PERIOD)

Because the Earth is curved, accelerating over a spherical surface would cause accelerometers to tilt and mistake gravity for acceleration, causing velocity errors to increase exponentially towards infinity:

- Schuler Pendulum: The IRS platform/computer is tuned to oscillate like a pendulum whose length equals the radius of the Earth ($R = 6,371 \text{ km}$). Its natural oscillation period is: $T = 2\pi \times \sqrt{R/g} = \text{EXACTLY } 84.4 \text{ MINUTES!}$
- Error Bounding: Schuler tuning forces velocity errors to oscillate harmlessly between positive and negative values every 84.4 minutes, preventing unbounded divergence!

3. IRS DRIFT & TRIPLE-MIX FMC INTEGRATION

- Bounded vs Unbounded Errors: Velocity and tilt errors oscillate with the 84.4 min Schuler cycle (bounded). Position errors creep forward with a steady DRIFT (unbounded, typically 1 to 2 NM per hour of flight).
- Triple Mix & Sensor Fusion: The Flight Management Computer (FMC) averages positions from 3 independent IRS units (Triple-Mix) and continuously updates the hybrid navigation solution using GPS and DME/DME updating.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the natural period of the SCHULER OSCILLATION in an Inertial Reference System (IRS)?

[A] 24 hours [B] EXACTLY 84.4 MINUTES [C] 60 minutes [D] 4 minutes

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Schuler period corresponds to an Earth-radius pendulum: $T = 2\pi \times \sqrt{R/g} = 84.4 \text{ minutes}$, bounding acceleration errors.

Q2: During stationary ground alignment of an IRS, what parameters can the system determine completely auto

[A] Only magnetic variation [B] TRUE NORTH and local LATITUDE (by sensing Earth's rotation rate and gravity) [C] Longitude and altitude [D] Runway heading

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Earth rotation vector (15°/h) provides True North and latitude = $\arcsin(\text{vertical component} / 15^\circ)$. Longitude has zero rotational signature and must be entered manually.

Q3: What optical physical effect is utilized by Ring Laser Gyros (RLG) to measure aircraft angular rotation?

[A] Photoelectric effect [B] THE SAGNAC EFFECT (frequency difference between counter-rotating laser beams) [C] Hall effect [D] Coriolis effect

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Rotation of the RLG alters path lengths of clockwise and counter-clockwise laser beams, generating an optical beat frequency proportional to turn rate.